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2027 CYCLE

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CUET UG

Biology

Reproductive Health

Official Syllabus Unit VI · Class 12 · CUET UG 2027 Cycle Notes

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Biology syllabus 2026, the current 2027 baseline, Unit VI)	Skills the questions test
Problems and strategies	Problems and strategies – amniocentesis;	Recalling what amniocentesis reads, and why its misuse is a punishable offence.
Population stabilisation and birth control	Population stabilisation and birth control – various methods of contraception;	Matching each contraceptive to its family and to how it actually works.
Medical termination of pregnancy	Medical termination of pregnancy (MTP);	Stating the legal windows and the safe-practice rules.
Sexually transmitted infections	Sexually transmitted infections (STIs);	Pairing each STI with its causative agent and its curable or treatable status.
Infertility	Infertility – Assisted reproductive technologies (IVF, ZIFT, GIFT, ICSI and IUI);	Matching each ART with what is transferred and where it is placed.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Biology paper maps strictly to NCERT Class 12, and Unit VI closes the reproduction block with applied biology: staying healthy, planning families, and the technologies that help when nature needs support.

What Reproductive Health Means

One household raises two children comfortably — school fees paid, doctor visited on time, parents calm. The next household has nine children and struggles for food, space and school uniform. The difference between the two homes is rarely luck. It is reproductive health: the ability to decide freely and responsibly how many children to have, when to have them, and to stay healthy while doing it. The **World Health Organisation (WHO)** — the UN body that sets global health standards — treats this as a basic part of health itself, and India has built national programmes around it since 1951.

- **The WHO definition:** reproductive health is **total well-being** in all matters of reproduction — **physical, emotional, behavioural and social** — not simply the absence of disease.
- **It is more than body health:** a person may have a perfectly healthy body yet carry fear, pressure or wrong information about sex and family life. WHO counts those troubles too.
- **Decision power:** every adult couple should be able to choose the size of their family and the spacing of children, without coercion.
- **Why society cares:** family planning keeps family size within what the family can feed and educate; **safe motherhood** keeps mothers alive through pregnancy and childbirth; **STI control** stops infections such as HIV from spreading.

India's programmes in one timeline

Year	What the country started
1951	India launches the world's first nation-wide **Family Planning Programme** — promoting small families and contraceptive use.
1970s–80s	Awareness campaigns widen: posters, radio, village health workers; contraceptive choices multiply.
1997	**RCH (Reproductive and Health Care) programmes** begin and continue onward — run under the national population policy, and widened to cover the child's health as well as the mother's.
Today	RCH ideas continue inside national health missions: antenatal check-ups, institutional deliveries, immunisation, contraception counselling.

- **RCH in one line:** RCH means **Reproductive and Health Care** — a programme that looks after the reproductive health of adults **and** the health of the children they bear.
- **The aim is stabilisation:** India's population policy aims at **population stabilisation** — a state where the population stops growing and levels off.
- **How awareness is spread:** schools, media and health workers all carry the same message — plan the family, protect the mother, prevent infection.

The tools that build awareness

- **Sex education in schools:** proper information about reproductive organs, adolescence, hygiene and safe behaviour — given **before** wrong information arrives from playgrounds and the internet.
- **Audio-visual media and posters:** short films, street plays and wall posters reach couples and students who may never open a health manual.
- **Fertile-window education:** teaching couples which days of the menstrual cycle carry the highest chance of pregnancy — knowledge used both to **avoid** and to **achieve** pregnancy.
- **Early medical help:** awareness campaigns urge couples with fertility problems to see a doctor in time instead of hiding for years.

EVERYDAY EXAMPLE — THE STEERING WHEEL

A car without a steering wheel goes wherever the road pushes it. **Reproductive health hands a couple the steering wheel of their family life:** when to start, how many children, how far apart. The WHO definition simply adds a service schedule — check the body (physical), the mind (emotional), the habits (behavioural) and the relationships (social). A car needs all four wheels; this definition needs all four well-beings.

Myths versus facts, in one line each

The myth you will hear	The fact to write in the exam
"If a couple has no child, the wife is at fault."	Infertility can sit with **either** partner — male, female, or both.
"Talking about reproductive health invites bad behaviour."	Correct education builds **responsible** behaviour; ignorance spreads infections and unsafe choices.
"Family planning means having no children."	It means **planned** children — the number and spacing a couple chooses.

What each well-being covers

Well-being	What it means here	One everyday example
Physical	Reproductive organs, hormones and cycles working normally; freedom from infection.	Regular menstrual cycles; an infection treated in time.
Emotional	Feeling safe and free from fear, shame or pressure about sex and family life.	A couple discussing family size calmly, without coercion.
Behavioural	Habits that protect health — the actions people actually take.	Using condoms; keeping antenatal check-up appointments.
Social	Equal respect for women; honest community support.	A village accepting a couple's choice of a small family.

- **A shared job:** governments build the services, schools supply the facts, families use both — reproductive health is not one person's burden.
- **The link forward:** every section that follows — contraception, MTP, STI control, infertility care — is simply one arm of the same RCH umbrella.

Amniocentesis and Population Strategies

Doctors can open the first page of a baby's health record months before birth. A needle, a few drops of the fluid that surrounds the foetus, and a laboratory microscope can reveal the baby's chromosomes — the thread-like structures that carry genes. This test is **amniocentesis**, and it is

one of the most examined topics of this chapter, because it has two faces: a life-saving diagnostic use and a banned, shameful misuse.

- **The test defined:** amniocentesis is a procedure where a small sample of **amniotic fluid** — the liquid filling the sac around the foetus in the womb — is withdrawn and studied.
- **Why the fluid helps:** the fluid floats with cells shed by the foetus, and those cells carry the **foetal chromosomes** — the full genetic instruction set.
- **Chromosomal disorders found:** counting and examining the chromosomes reveals disorders such as **Down's syndrome**, where an extra copy of chromosome 21 (47 chromosomes in total instead of 46) changes development.
- **Metabolic disorders found:** the foetal cells can also be tested for working **enzymes** — proteins that speed up the body's chemical reactions. A missing enzyme points to an inherited **metabolic disorder**, in which the body cannot process some substance properly.

Two faces of the same test

The legal use	The banned use
Reading foetal chromosomes to detect chromosomal disorders such as Down's syndrome.	Determining the **sex** of the foetus.
Checking foetal enzymes to detect metabolic disorders.	Ending pregnancies of female foetuses — **female foeticide**.
Done in registered facilities for genuine medical reasons.	Punishable by law, with heavy penalties for doctors and parents.

- **The law:** the **Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994** (known in short as the PCPNDT Act) bans prenatal sex determination and sex-selective abortions.
- **Why the law exists:** misuse for sex determination caused millions of missing girls, skewing the child sex ratio — a social wound, not just a medical one.
- **The exam angle:** a classic question frame pairs "amniocentesis detects" with "amniocentesis must never be used for" — memorise both columns above.

WORKED EXAMPLE — READING AN AMNIOCENTESIS REPORT

A laboratory receives amniotic fluid from a 17-week pregnancy and reports: **47 chromosomes, with three copies of chromosome 21; enzyme panel normal.**

- **Step 1 — count:** a normal human count is 46. This report shows 47 — one extra chromosome.
- **Step 2 — identify:** the extra chromosome is chromosome 21, present in three copies (trisomy 21).
- **Step 3 — conclude:** trisomy 21 is **Down's syndrome** — a chromosomal disorder, which is exactly what amniocentesis is designed to detect.
- **Step 4 — the guardrail:** the report says nothing about sex, and no doctor may reveal the foetal sex under the PCPNDT Act, 1994.

Population stabilisation: why and how

- **The pressure:** every extra mouth needs food, water, housing, schooling, healthcare and eventually a job. A population growing faster than resources pushes families and the country into strain.
- **The goal: population stabilisation** — bringing growth down until the population levels off at a sustainable size.
- **Incentives:** governments reward small families — easier loans, health benefits and welfare schemes linked to adopting family planning.
- **Education:** educated couples marry later, plan children better and adopt contraception more readily; the legal marriage ages (18 for women, 21 for men) exist partly for this reason.
- **Awareness plus access:** posters, media campaigns and free contraceptive services work together — knowing about methods is useless without being able to get them.
- **The multiplier effect:** one aware couple teaches the next; that is why RCH programmes spend as much effort on **education** as on supplies.

Four stabilisation strategies, judged

Strategy	How it slows growth	One caution
Incentives for small families	Rewards — easier loans, welfare benefits — make the two-child family attractive.	Must encourage, never punish the poor.
Education, especially of women	Educated women marry later, plan children better and use contraception more readily.	Takes a generation to show its full effect.
Raising the marriage age	Fewer child-bearing years per couple; the law sets 18 for women, 21 for men.	A law on paper needs enforcement to matter.
Easy contraceptive access	Free, nearby choices remove the excuse of distance and cost.	Services must stay confidential and respectful.

- **Stabilisation has momentum:** even after birth rates fall, the total population keeps rising for years, because so many couples are young and yet to start families.
- **The syllabus pairing:** this unit puts amniocentesis and population strategies in one line — one is a diagnostic test that was misused, the other a nation that must be steered.
- **The honest scale:** awareness changed faster than numbers — behaviour change is slow, which is exactly why education gets the long budget in every population policy.

Contraception I — Natural and Barrier

A couple wants to delay their first child for two years. Another has two children and wants none more. Both need the same thing: methods that prevent pregnancy **without harming health**. That is the whole idea of **contraception** — the deliberate prevention of pregnancy. The simplest

families of methods need no hormones and no surgery: one family reads the body's calendar, the other puts a physical wall between sperm and egg.

- **The principle:** a good contraceptive method should be safe, affordable, easy to use and free of long-term side effects — no method is perfect, so couples pick what suits their health and plans.
- **Natural methods** rely on timing or simple behaviour — zero cost, but the highest failure rate of all families.
- **Barrier methods** physically stop the sperm from reaching the egg — cheap, reliable when used properly, and the only family that also blocks infections.

The natural (behavioural) methods

- **Periodic abstinence:** the couple avoids intercourse during the fertile window of the menstrual cycle — roughly **days 10 to 17** of a 28-day cycle — because ovulation (the release of an egg from the ovary) happens around day 14.
- **Withdrawal (coitus interruptus):** the male withdraws before ejaculation, the release of semen, so sperm do not enter the female tract. Discipline decides success, and it often fails.
- **Lactational amenorrhoea:** intense, **exclusive** breastfeeding keeps the mother's ovulation switched off, so menstrual periods stop (**amenorrhoea**) and pregnancy is unlikely — but only for about **six months**, and only while feeding is frequent and no other milk or food is given.
- **The honest warning:** natural methods carry the **highest failure rate** among all contraceptive families, because they depend on perfect timing and perfect self-control.

WORKED EXAMPLE — FINDING THE FERTILE WINDOW

A woman has a regular 28-day cycle. Her last period began on the 1st of the month. The couple wants to practise periodic abstinence.

- **Step 1 — count the days:** cycle days are counted from the **first day of menstrual flow**, so the 1st is day 1.
- **Step 2 — locate ovulation:** in a 28-day cycle ovulation falls near **day 14**, when a mature egg leaves the ovary.
- **Step 3 — add margins:** sperm can survive in the female tract for about **3 to 5 days**, and the egg lives about **1 day**. So pregnancy is possible from about day 10 to day 17.
- **Step 4 — the rule:** avoid intercourse from **day 10 to day 17** — the fertile window this method treats as unsafe.

The barrier methods

Method	How it works	Extra benefit / limit
Male condom	A thin rubber sheath rolled over the penis; it collects semen and blocks sperm entry.	Also blocks STIs — cheap and easily available; must be used every time.

Method	How it works	Extra benefit / limit
Female condom	A pouch lining the female tract; blocks sperm before they reach the cervix.	Woman controls its use; also blocks STIs; slightly costlier.
Diaphragm	A flexible rubber dome fitted over the cervix — the mouth of the uterus.	Reusable; must be fitted correctly and left in place after intercourse.
Cervical cap	A smaller cap covering only the cervix.	Same idea as the diaphragm; needs a trained fitter for the right size.
Spermicides	Creams and jellies that kill sperm on contact.	Used **with** diaphragms or caps; never reliable alone.

EVERYDAY EXAMPLE — THE STADIUM GATE

Barrier methods are the security gate of the reproductive tract: the match (fertilisation) can only happen if sperm get past the gate. A **condom** is a full gate with a collection bin attached, a **diaphragm** or **cervical cap** locks the entry door (the cervix), and a **spermicide** is the guard who removes any stragglers. Natural methods do something different — they simply keep the star player (the egg) off the field on match days.

Natural vs barrier at a glance

Point	Natural methods	Barrier methods
How pregnancy is prevented	Timing or withdrawal — no device, no medicine.	A physical wall stops sperm from meeting the egg.
Cost	Free.	Small, but repeated with every use.
Protection from STIs	None.	Yes — condoms block infections as well as sperm.
Failure rate	Highest of all families.	Low when used correctly, every single time.
Best suited for	Couples with regular cycles and strong discipline.	Couples who want simple, sure protection.

Four quick judges

Claim	Verdict	Why
"Breastfeeding is a foolproof contraceptive."	False	Lactational amenorrhoea works only about six months, and only with exclusive, frequent feeding.
"Withdrawal is a reliable method."	False	Fluid released before ejaculation can carry sperm; self-control fails often.
"A diaphragm works better when paired with a spermicide."	True	The dome blocks the cervix; the cream kills any sperm that slip past.
"Natural methods cost nothing, so they suit every couple."	False	Cost is zero, but the high failure rate defeats couples who badly need to avoid pregnancy.

Contraception II — IUDs, Pills, Sterilisation

Barrier methods work every single time they are used; devices, pills and surgery work for months or years after a **single step**. Intrauterine devices sit inside the womb for years; pills reset the body's hormone signals each day; sterilisation closes the transport tubes once and for all. This is the most "matching-heavy" part of the chapter — examiners pair each device with its class and its action, so learn the families precisely.

The three classes of IUDs

An **IUD (intrauterine device)** is a small device placed inside the uterus, where it can stay for years. It is inserted by a doctor and works round the clock — the couple does nothing.

Class	Named devices	How the class works
Non-medicated IUDs	Lippes loop	An inert plastic loop; its presence in the womb makes white blood cells engulf sperms (phagocytosis), so few sperms survive to fertilise the egg.
Copper-releasing IUDs	Cu-T, Cu-7, Multiload 375	Copper ions released by the device **suppress sperm motility** (their ability to swim) and their fertilising capacity.
Hormone-releasing IUDs	Progestasert, LNG-20	They release hormones that make the uterus unsuitable for **implantation** — the embedding of the early embryo; some also stop ovulation.

- **IUD in one line:** a long-acting device inside the womb — but remember, an IUD does **not** protect against STIs.

Pills: daily, weekly, emergency

Pill type	What it contains	How it is used
Oral contraceptive pills	Norethindrone (a progestogen) plus ethinylestradiol (an oestrogen) — two hormones	Taken daily; the hormone dose **inhibits ovulation**, so no egg is released to be fertilised.
Saheli	Centchroman — a **non-steroidal** pill	Taken **once a week**; hardly any side effects; developed in India — a pride of Indian pharmacology.
Emergency contraceptive pills	High hormone doses	Used within about **72 hours** after unprotected intercourse — an emergency rescue, **not** a routine method.

- **Why hormonal pills work:** a steady supply of outside hormones fools the brain's cycle control, so the ovary holds its egg back.
- **Saheli's special place:** most pills are **steroidal** (hormone-based); Saheli is not, which is why its few side effects made it famous.

Sterilisation: vasectomy vs tubectomy

Point	Vasectomy (male)	Tubectomy (female)
Who	The male partner.	The female partner.
What is cut and tied	The **vas deferens** — the tube that carries sperm from each testis.	The **fallopian tubes** — the tubes that carry the egg from ovary to uterus.
Effect	Semen is released without sperm — ejaculation still feels normal.	Eggs cannot meet any sperm — pregnancy becomes impossible.
Nature of the step	A quick, near-painless minor operation.	A slightly longer abdominal operation.
Reversibility	Rejoining the tubes is possible but **not guaranteed** — treat it as permanent.	Same caution: reversal surgery exists but success is not assured.

- **The matching trap:** vasectomy and tubectomy sound similar but belong to different partners — the **vas** deferens is the male tube, and **tubectomy** cuts the female tubes.
- **Sterilisation in one line:** a terminal method for couples who have completed their family — highly effective, but plan it as permanent.

Match the device to its class

Device or pill	Family	One-line action
Lippes loop	Non-medicated IUD	The loop's presence makes white blood cells in the womb engulf sperms.
Cu-T	Copper-releasing IUD	Copper ions suppress sperm motility and fertilising capacity.
LNG-20	Hormone-releasing IUD	Hormones make the uterus unfit for implantation; ovulation may stop.
Norethindrone + ethinylestradiol	Steroidal daily pill	The hormone dose inhibits ovulation.
Saheli (centchroman)	Non-steroidal weekly pill	Once a week, few side effects — made in India.
Vas deferens cut and tied	Sterilisation — vasectomy	Semen is released without sperm.

- **The drill habit:** cover the middle column and name the family for each device — this is exactly the matching style CUET questions use.
- **One family is missing above:** tubectomy — the female sterilisation, which cuts and ties the fallopian tubes.

Medical Termination of Pregnancy

No couple plans for a pregnancy to end. Yet some pregnancies cannot continue safely — the mother's health is in danger, the foetus has a serious defect, or the pregnancy followed a crime such as rape. When a pregnancy is ended **deliberately**, by choice and by medical procedure, it is called **MTP (medical termination of pregnancy)** — an intentional or voluntary abortion. India made this safe and legal nearly fifty years ago, and the rules form an exam-table students must memorise precisely.

- **The definition:** MTP is the intentional or voluntary termination of pregnancy before full term, performed as a medical procedure.
- **The law:** MTP is legal in India under the **Medical Termination of Pregnancy Act, 1971**, amended in 2017 and again in 2021 to widen safe access.
- **Not the same as miscarriage:** a miscarriage is a **spontaneous** abortion — the body ends the pregnancy on its own; MTP is a **chosen** medical act.

The legal windows, week by week

Stage of pregnancy	What the law allows	Who certifies it
Up to 12 weeks (first trimester)	MTP may be performed — this is the safest and most common window.	One registered medical practitioner.
12 to 24 weeks (second trimester)	MTP may still be performed, with extra care and two medical opinions.	Opinion of registered medical practitioners.
Beyond 24 weeks	Only in narrow special cases — for example, severe foetal abnormality or a serious risk to the mother's life (2021 amendment).	A medical board of specialists.

- **Why the first 12 weeks matter:** the earlier the termination, the simpler the procedure; the risk rises steadily after the first trimester.
- **The methods in one line:** early pregnancies are ended by **suction or vacuum aspiration** (gentle suction empties the uterus) or by **medicines**; later ones may need surgery.
- **Qualified hands only:** MTP must be done by a qualified, registered doctor in an approved facility — never by unqualified practitioners.

The cost of unsafe abortion

- **The danger list:** unsafe abortions — done by untrained hands in unhygienic conditions — cause heavy bleeding, severe infection and even death.
- **The lasting damage:** an unsafe procedure can scar the reproductive tract and leave a woman **infertile** for life.
- **The social point:** legal, safe MTP exists precisely so that no woman has to risk her life in secret; restricting access does not stop MTPs, it only makes them dangerous.
- **The partner's role:** a safe MTP also needs a supportive family — secrecy and stigma are what push women toward unsafe hands.
- **The numbers to memorise:** 12 and 24 — the two week-marks that carry every MTP question, tied to the words "first trimester" and "second trimester".
- **Who may certify:** a **registered medical practitioner** — a doctor with a recognised qualification, entered on a state medical register, working at an approved facility.
- **The counselling rule:** good doctors pair MTP with counselling and contraception advice, so the next pregnancy is a planned one.

Worked Example — Applying the legal windows

- **Case A — a woman is 8 weeks pregnant and requests termination:** this is the first trimester. One registered medical practitioner can certify it, and the procedure is at its safest — suction aspiration or medicines.

- **Case B — a foetal abnormality is suspected at 15 weeks:** the second trimester window (12–24 weeks) still allows MTP, with the opinion of registered medical practitioners, and carries more procedural risk.
- **Case C — a request arrives at 27 weeks:** beyond 24 weeks, only narrow special cases qualify — a medical board weighs severe foetal abnormality or danger to the mother's life (the 2021 amendment). An ordinary request does not qualify.
- **Case D — a pregnancy that followed rape, at 11 weeks:** the first-trimester window applies with one practitioner's opinion; the law names such cases specifically and keeps the procedure confidential.

MTP versus miscarriage

Point	MTP	Miscarriage
Cause	Chosen deliberately, by the couple with doctors.	Happens on its own — the body ends the pregnancy.
Medical term	Induced (voluntary) abortion.	Spontaneous abortion.
Law involved	Regulated by the MTP Act, 1971 and its amendments.	No law applies — it is a medical event, not a choice.
Doctor's role	Performs the procedure in an approved facility.	Provides care and monitoring after the event.
Aftercare	Contraception counselling follows, so the next pregnancy is planned.	Rest, monitoring and emotional support follow.

Sexually Transmitted Infections

Infections travel along more routes than one. Some ride on coughs and dirty water; a stubborn group travels through **sexual contact**. These are the **STIs (sexually transmitted infections)** — also called venereal diseases. They are among the most common infectious diseases worldwide, and this chapter treats them matter-of-factly: know the agent, know whether it is curable, know how to prevent it. Health education, not embarrassment, is the answer.

- **The definition:** STIs are infections transmitted through sexual contact, from an infected partner to an uninfected one.
- **The early signs:** discharge from the genitals, itching, sores and pain in the pelvic region — but many infections show **no clear symptoms at first**, which is exactly why they spread silently.
- **The hidden damage:** untreated STIs can climb upward and cause **pelvic inflammatory disease** (long-term infection of the female tract), infertility, or complications in pregnancy such as stillbirth.
- **The silent-carrier rule:** a person can look and feel perfectly healthy and still carry and pass on an STI — so risk behaviour, not appearance, decides safety.

The complete roster, agent by agent

STI	Causative agent	Type	One key fact
Gonorrhoea	**Neisseria gonorrhoeae** (bacterium)	Bacterial	Causes burning urination and discharge; curable with antibiotics.
Syphilis	**Treponema pallidum** (bacterium)	Bacterial	Starts as a painless sore; curable if treated early.
Chlamydiasis	**Chlamydia** (bacterium)	Bacterial	Often symptomless — a classic silent spreader.
Genital herpes	**Herpes simplex virus**	Viral	Painful blisters that return; treatable, not curable.
Genital warts	**Human papilloma virus (HPV)**	Viral	Warts on the genitals; linked to cancers of the tract.
Trichomoniasis	**Trichomonas** (protozoan)	Protozoan	Causes frothy discharge; treatable with medicines.
Hepatitis-B	Hepatitis-B virus	Viral	Also spreads by shared needles and infected blood; damages the liver.
HIV	Human Immunodeficiency Virus	Viral	Attacks the immune system and leads to AIDS; also blood-borne.

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- **The curable half:** bacterial STIs — gonorrhoea, syphilis, chlamydiasis — are **completely curable** with antibiotics when caught early.
- **The stubborn half:** viral STIs — herpes, warts, hepatitis-B, HIV — can be **treated and controlled but never fully cured**; the virus stays in the body.
- **The extra routes:** hepatitis-B and HIV also spread through transfusion of infected blood and sharing of contaminated needles, not only through sex.

Prevention: the working checklist

- **Condoms:** the single habit that blocks both pregnancy and STIs.
- **Fewer, safer choices:** avoiding sex with multiple partners cuts risk sharply; one infected partner can infect every partner that follows.
- **Hygiene and honesty:** personal cleanliness and open discussion before marriage protect couples — silence protects nobody.
- **Screened blood:** accept blood only from licensed, tested blood banks; insist on new, sterile needles.

Worked Example — Reading a symptom report

A 24-year-old reports burning urination and a yellow discharge, three weeks after a new partner. Walk the diagnosis the way a doctor would.

- **Step 1 — read the signs:** burning urination plus genital discharge point to a bacterial STI — gonorrhoea or chlamydiasis are the first suspects.
- **Step 2 — confirm the agent:** a swab test identifies **Neisseria gonorrhoeae** or **Chlamydia** under the microscope and in the lab.
- **Step 3 — treat early:** antibiotics **cure** bacterial STIs completely when started in time.
- **Step 4 — treat the pair:** the partner is tested and treated too, or the infection simply travels back and forth between the two.
- **Step 5 — the caution:** had the agent been a virus — herpes or HIV — the same report would end in treatment and control, never in cure.

When an STI goes untreated

Complication	What happens in the body
Pelvic inflammatory disease	The infection climbs into the uterus and tubes, causing long-term pelvic pain.
Infertility	Scarred tubes block the egg and sperm from meeting.
Pregnancy complications	Stillbirth becomes possible, or the infection passes to the baby.
Cancer risk	Long-lasting HPV infection is linked to cancers of the reproductive tract.

Infertility and ART

Some couples do everything right, yet month after month no pregnancy begins. Doctors call it **infertility** when a couple fails to conceive after one to two years of regular, unprotected attempts. It is more common than people think, and — this is the sentence every student should carry home — **the cause can sit with either partner**, the man or the woman, equally often. Modern medicine has built a toolkit for exactly this problem, and this section matches each tool to its job.

- **The causes:** they range from physical problems (blocked tubes) to congenital ones (present from birth), diseases, hormonal imbalances, certain drugs, and even stress.
- **The diagnosis rule: both** partners are examined — blaming the woman by default is a myth the WHO definition has buried.
- **The simpler steps first: hormonal treatment** can correct imbalances, and **artificial insemination** — collecting semen and placing it into the female tract by a doctor — can succeed without anything elaborate.

When nature needs a helper: the ART toolkit

When simpler steps fail, couples turn to **ART (assisted reproductive technologies)** — special techniques that handle the egg, the sperm or the early embryo in the laboratory. Learn each method

as three questions: **what is collected or transferred, where does it go, and when does it fit?**

- **IVF — In Vitro Fertilisation:** "in vitro" means in glass. Eggs and sperm meet **outside** the body, and the embryo is transferred into the uterus once it has **8 or more blastomeres** (cells). Every famous "test-tube baby" began this way.
- **ZIFT — Zygote Intrafallopian Transfer:** the **zygote** (the freshly fertilised single cell) or an early embryo of **up to 8 blastomeres** is placed into the **fallopian tube**, and travels to the uterus on its own.
- **GIFT — Gamete Intrafallopian Transfer:** no lab fertilisation at all — the **gametes** (an egg and sperm) are placed into the fallopian tube, and fertilisation happens **inside the body**.
- **ICSI — Intra Cytoplasmic Sperm Injection:** one sperm is picked up with a fine needle and **injected directly into the cytoplasm of the egg** — the answer when sperm cannot enter the egg by themselves.
- **IUI — Intra Uterine Insemination:** prepared (washed, concentrated) semen is placed **into the uterus** — the simplest ART, used for low sperm count or poor sperm quality.

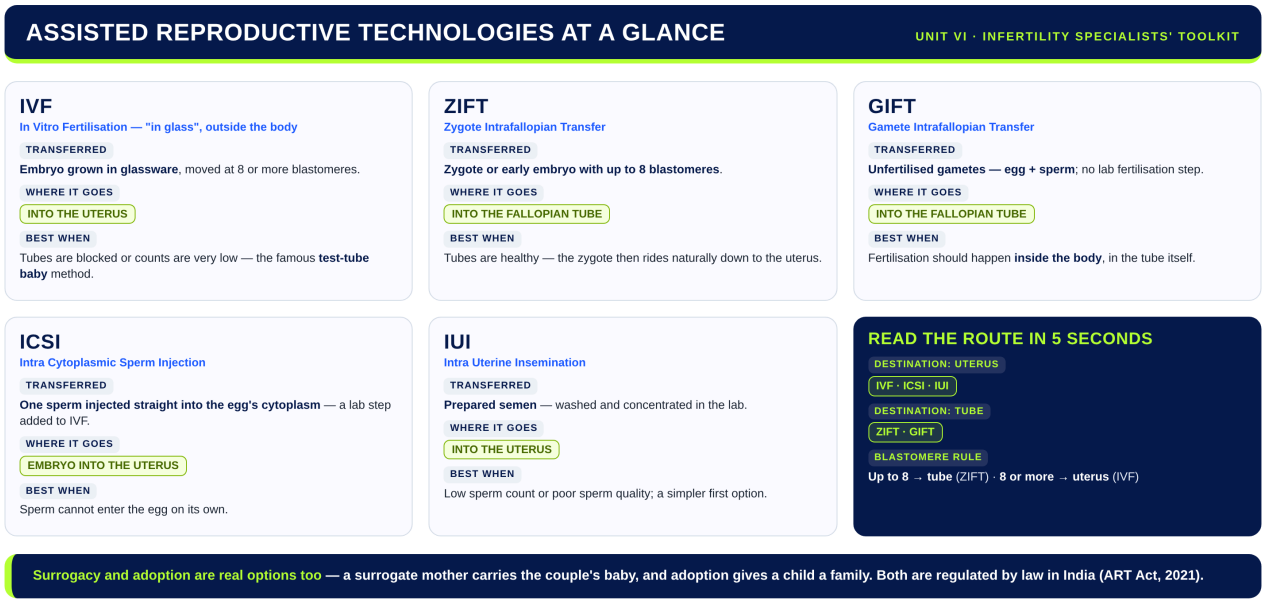


Figure 1: The five assisted reproductive technologies compared — what is transferred, where it goes, and when each fits best.

WORKED EXAMPLE — CHOOSING THE RIGHT ART

Match each couple to the technology a specialist would consider first.

- **Couple A — the woman's fallopian tubes are irreversibly blocked:** sperm and egg cannot meet in the tube at all, so fertilisation must happen outside — **IVF**, with the 8-or-more-blastomere embryo placed in the uterus.
- **Couple B — the man has a low sperm count:** the simplest boost first — **IUI**, placing prepared semen straight into the uterus.
- **Couple C — the sperm cannot penetrate the egg on their own:** no amount of proximity will help; a single sperm must be injected into the egg — **ICSI**.

- **Couple D — healthy tubes, and doctors want the embryo to develop in the tube's natural setting:** transfer the early stage into the tube — **ZIFT** (up to 8 blastomeres) or **GIFT** (gametes, fertilisation inside).

The complete answers beyond technology

- **Counselling:** both partners carry stress and guilt; professional counselling supports couples through every stage of treatment.
- **Surrogacy:** a **surrogate mother** carries the couple's baby in her own uterus when the woman cannot — an arrangement India now regulates by law.
- **Adoption:** a completely valid way to build a family, and the chapter's final message — a family is built by love, not only by biology.
- **The regulation one-liner:** India's **ART Act, 2021** registers clinics, sets ethical standards and protects the rights of everyone involved.

Before ART: the simpler rungs

Step	What happens	Fits when
Counselling	A professional eases stress and guilt and guides the couple's decisions.	Always — alongside every other step.
Hormonal treatment	Corrects hormone imbalances so ovulation restarts.	Blood tests show the imbalance.
Artificial insemination	A doctor collects semen and places it into the female tract.	Sperm count or quality is mildly low.

- **The ladder rule:** doctors climb from the simplest rung upward — elaborate laboratory work comes last, after simple steps have had their chance.
- **Either-partner rule again:** the ladder is climbed by the couple, not by the woman alone.

Key Facts & Terms Cheatsheet: the Whole Unit in One Table

KEY FACTS & TERMS CHEATSHEET

Term / fact	Meaning	Example / number
Reproductive health (WHO)	Total well-being — physical, emotional, behavioural, social — in all reproduction matters.	Not just "no disease".
India's programmes	First family-planning programme, then RCH (Reproductive and Health Care) programmes.	1951; RCH from 1997.

Term / fact	Meaning	Example / number
Amniocentesis	Sampling amniotic fluid to read foetal chromosomes and enzymes.	Detects Down's syndrome; metabolic checks.
Amniocentesis ban	Prenatal sex determination and sex-selective abortion.	PCPNDT Act, 1994.
Natural methods	Periodic abstinence, withdrawal (coitus interruptus), lactational amenorrhoea.	Avoid days 10–17; feeding up to about 6 months.
Barrier list	Condoms (male and female), diaphragm, cervical cap, spermicides.	Condoms also block STIs.
IUD classes	Non-medicated, copper-releasing, hormone-releasing.	Lippes loop; Cu-T, Cu-7, Multiload 375; Progestasert, LNG-20.
Copper action	Copper ions suppress sperm motility and fertilising capacity.	Cu-T releases Cu ions.
Hormone-IUD action	Make the uterus unsuitable for implantation; some stop ovulation.	LNG-20, Progestasert.
Regular pills	Norethindrone + ethinylestradiol — inhibit ovulation.	Taken daily.
Saheli	Centchroman — non-steroidal, few side effects.	Once a week; made in India.
Emergency pills	Rescue dose after unprotected intercourse.	Within about 72 hours — never routine.
Vasectomy	Vas deferens of the male cut and tied.	Quick minor operation.
Tubectomy	Fallopian tubes of the female cut and tied.	Treat as permanent.
MTP windows	Up to 12 weeks safest; 12–24 weeks with practitioners' opinion; beyond 24 only special cases.	MTP Act, 1971; amended 2021.
Bacterial STIs	Gonorrhoea — **Neisseria gonorrhoeae**; syphilis — **Treponema pallidum**; chlamydiasis — **Chlamydia**.	Curable with antibiotics.

Term / fact	Meaning	Example / number
Viral STIs	Genital herpes — herpes simplex virus; warts — HPV; hepatitis-B; HIV.	Treatable, not curable.
Protozoan STI	Trichomoniasis — **Trichomonas**.	Treatable with medicines.
IVF	Fertilisation outside the body; embryo into the uterus.	At 8 or more blastomeres; "test-tube baby".
ZIFT	Zygote or early embryo placed into the fallopian tube.	Up to 8 blastomeres.
GIFT	Gametes (egg + sperm) placed into the tube; fertilisation inside the body.	No lab fertilisation.
ICSI	One sperm injected into the egg's cytoplasm.	For sperm that cannot enter the egg.
IUI	Prepared semen placed into the uterus.	For low sperm count or quality.

Carry these numbers: fertile window days 10–17 · exclusive breastfeeding about 6 months · emergency pills within 72 hours · MTP windows 12 and 12–24 weeks · embryo route rule: up to 8 blastomeres into the tube, 8 or more into the uterus · normal chromosome count 46.

Common Errors and Special Cases

COMMON ERRORS & SPECIAL CASES

The mistake students make	What is right	Why / the trap
Saying pills and IUDs also protect against STIs	Only **barrier** methods — condoms — block both pregnancy and STIs.	Pills and IUDs act on hormones or the womb; no virus meets a wall.
Calling Cu-T a hormone-releasing IUD	Cu-T, Cu-7 and Multiload 375 are **copper-releasing**; Progestasert and LNG-20 are hormone-releasing.	The metal in the name marks the copper family.

The mistake students make	What is right	Why / the trap
Grouping Saheli with steroidal pills	Saheli is **non-steroidal** (centchroman), taken once a week.	Most other pills — Norethindrone + ethinylestradiol — are steroidal.
Placing the ZIFT embryo in the uterus	ZIFT transfers into the **fallopian tube**; IVF transfers 8-or-more-blastomere embryos into the **uterus**.	The blastomere count decides the route: up to 8 — tube; 8 or more — uterus.
Describing GIFT as embryo transfer	GIFT transfers **gametes** — an egg and sperm.	Fertilisation happens inside the tube, not in a dish.
Swapping vasectomy and tubectomy	Vasectomy cuts the **vas deferens** (male); tubectomy cuts the **fallopian tubes** (female).	"Vas" belongs to the male tube — do not swap sides.
Treating viral STIs with antibiotics and expecting cure	Antibiotics cure **bacterial** STIs only; viral STIs are treatable but not curable.	Herpes and HIV stay in the body for life.

Three checking habits that beat every trap above

- **Name the family, not just the device:** every contraceptive question is a family question — say "copper-releasing IUD" in your head, never just "Cu-T".
- **Route before method:** for ART, decide the destination first — uterus or tube — then check the blastomere count and the cargo (embryo or gametes).
- **Cure or control:** for STIs, make the bacterial-versus-viral split first; everything else — agent, treatment, prevention — follows from it.
- **Window before answer:** MTP and emergency pills are both questions of weeks and hours — memorise the numbers (12, 12–24, 72) and attach them to the right method.
- **Whose body?** vasectomy and tubectomy are mirror procedures for two partners; before answering, say the partner's name in your head.

Special cases worth a second look

Situation	What applies
Pregnancy beyond 24 weeks with a severe foetal abnormality	Only a medical board of specialists can permit termination — the 2021 amendment's narrow lane.

Situation	What applies
A condom broke during the fertile window	Emergency pills within 72 hours are the rescue step — a one-time measure, never a routine method.
Both partners carry a curable STI	Both are treated together and finish the course — treating one alone only restarts the cycle.
A woman who cannot carry a pregnancy at all	Surrogacy or adoption — both lawful, both regulated, both complete answers.

The Whole Chapter on One Page

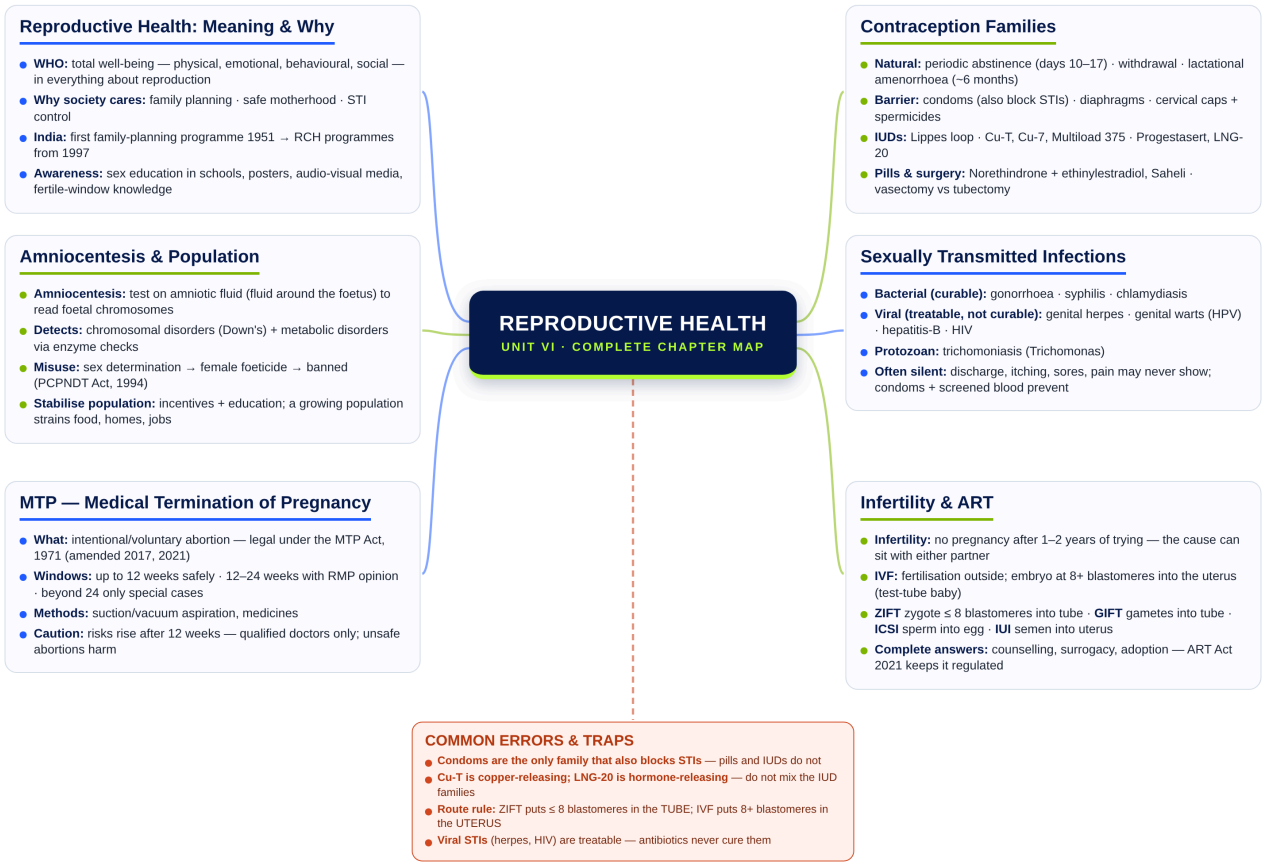


Figure 2: Chapter mindmap — every Unit VI idea of this chapter and the four traps that catch most students.

Practice Questions

Q1. A couple avoids intercourse only between days 10 and 17 of the woman's regular 28-day cycle. Which method are they practising?

(A) Lactational amenorrhoea
(B) Withdrawal
(C) Periodic abstinence
(D) Coitus interruptus

Q2. Cu-T, Cu-7 and Multiload 375 belong to which class of IUDs?

(A) Non-medicated
(B) Copper-releasing
(C) Hormone-releasing
(D) Steroidal oral pills

Q3. Which statement about Saheli is correct?

(A) It is a steroidal pill taken daily
(B) It must be taken within 72 hours of intercourse
(C) It is a non-steroidal pill taken once a week
(D) It protects against sexually transmitted infections

Q4. In IVF, the embryo is transferred into the uterus when it has reached:

(A) the gamete stage
(B) the zygote stage
(C) up to 8 blastomeres
(D) 8 or more blastomeres

Q5. In ZIFT, the zygote or early embryo of up to 8 blastomeres is transferred into the:

(A) uterus
(B) fallopian tube
(C) cervix
(D) ovary

Q6. Which technology places unfertilised gametes into the fallopian tube so that fertilisation happens inside the body?

(A) IUI
(B) ICSI
(C) GIFT
(D) ZIFT

Q7. A single sperm is injected directly into the cytoplasm of an egg in:

(A) ICSI
(B) GIFT
(C) ZIFT
(D) IUI

Q8. Which of these STIs is caused by a bacterium and is curable with antibiotics?

(A) Genital herpes
(B) Hepatitis-B
(C) AIDS
(D) Gonorrhoea

Q9. Under the MTP Act, a pregnancy can be safely terminated up to how many weeks during the first trimester?

(A) 4 weeks
(B) 12 weeks
(C) 24 weeks for any request
(D) 36 weeks

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Q10. Amniocentesis is legally banned in India when it is used for:

(A) determining the sex of the foetus

(B) detecting chromosomal disorders such as Down's syndrome

(C) checking foetal enzymes for metabolic disorders

(D) confirming foetal growth

Solutions

Sol. Q1. Days 10 to 17 cover ovulation (near day 14) plus the survival margins of sperm (3–5 days) and the egg (1 day). Avoiding this fertile window is **periodic abstinence**. Withdrawal and coitus interruptus are two names for the same method, and lactational amenorrhoea depends on breastfeeding.

Answer: (C)

Sol. Q2. The copper in each name marks the **copper-releasing** family: copper ions suppress sperm motility and fertilising capacity. Lippes loop is the non-medicated device, while Progestasert and LNG-20 are the hormone-releasing IUDs.

Answer: (B)

Sol. Q3. Saheli contains **centchroman**, a **non-steroidal** pill taken **once a week** with hardly any side effects — developed in India. It is not an emergency pill (72-hour rule) and no pill blocks STIs.

Answer: (C)

Sol. Q4. In IVF the embryo grows outside and is transferred into the uterus at **8 or more blastomeres**. Earlier stages belong to ZIFT (up to 8 blastomeres), and gametes are never the IVF transfer.

Answer: (D)

Sol. Q5. ZIFT stands for Zygote Intrafallopian Transfer — the zygote or an early embryo of **up to 8 blastomeres** is placed into the **fallopian tube**, from where it travels down to the uterus naturally.

Answer: (B)

Sol. Q6. GIFT — Gamete Intrafallopian Transfer — moves **unfertilised gametes** (egg + sperm) into the tube, so fertilisation happens **inside the body**. ZIFT transfers a zygote, ICSI injects a sperm into an egg, and IUI places semen in the uterus.

Answer: (C)

Sol. Q7. ICSI — Intra Cytoplasmic Sperm Injection — picks one sperm with a fine needle and injects it straight into the egg's **cytoplasm**. It is the rescue technology when sperm cannot penetrate the egg on their own.

Answer: (A)

Sol. Q8. Gonorrhoea is caused by the bacterium **Neisseria gonorrhoeae**, so antibiotics cure it. Genital herpes (herpes simplex virus), hepatitis-B and AIDS (HIV) are all **viral** — treatable but never fully curable.

Answer: (D)

Sol. Q9. The first-trimester window runs **up to 12 weeks**, when termination is safest and one registered medical practitioner can certify it. The 12–24-week window needs practitioners' opinions, and beyond 24 weeks only narrow special cases qualify.

Answer: (B)

Sol. Q10. The **Pre-Conception and Pre-Natal Diagnostic Techniques Act, 1994** bans prenatal sex determination and sex-selective abortions, because misuse led to female foeticide. The diagnostic uses — chromosomal and metabolic testing — remain legal.

Answer: (A)

The 60-second final drill

Recall prompt	Answer to say out loud
Which is the only contraceptive family that also blocks STIs?	Barrier methods — above all, condoms.
The three IUD classes in order, with one device each?	Non-medicated (Lippes loop), copper-releasing (Cu-T), hormone-releasing (LNG-20).
The pill hormones that inhibit ovulation?	Norethindrone + ethinylestradiol; Saheli is the non-steroidal weekly pill.
The MTP windows?	Up to 12 weeks; 12–24 weeks with practitioners' opinion; beyond 24, special cases only.
Two bacterial STIs with their agents?	Gonorrhoea — **Neisseria gonorrhoeae**; syphilis — **Treponema pallidum**.
The ART route rule in one line?	Up to 8 blastomeres into the tube (ZIFT); 8 or more into the uterus (IVF); gametes into the tube (GIFT).
The amniocentesis law and year?	PCPNDT Act, 1994 — bans prenatal sex determination.

Recall prompt	Answer to say out loud
The three natural methods, in order?	Periodic abstinence, withdrawal (coitus interruptus), lactational amenorrhoea.
Spermicides are always paired with?	Diaphragms and cervical caps — never trusted alone.
Which device's ions stop sperm from swimming?	Cu-T — copper-releasing IUD.
The two methods of MTP?	Suction or vacuum aspiration, and medicines.
MTP versus miscarriage in one line?	MTP is chosen and induced; miscarriage is spontaneous.
Beyond 24 weeks of pregnancy — what still applies?	Only special cases, cleared by a medical board.
Which STI does HPV cause, and its type?	Genital warts — viral, treatable but not curable.
A zygote of up to 8 blastomeres goes where?	The fallopian tube — that is ZIFT.
Which ART needs healthy tubes for the zygote to ride down?	ZIFT — the tube does the transporting naturally.
The bacteria behind syphilis?	**Treponema pallidum** — a curable bacterial STI.
Which daily pill is steroidal, and which weekly pill is not?	Daily: Norethindrone + ethinylestradiol. Weekly: Saheli — non-steroidal.
The WHO word that covers body, mind, behaviour and society?	Total well-being — the core of the reproductive-health definition.
The year of India's first family-planning programme?	1951 — RCH programmes followed from 1997.
The three complete answers for an infertile couple?	Counselling, surrogacy, adoption — under the ART Act, 2021.

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